

Runze Liu

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EDUCATION

Pennsylvania State University	2022 - Present
• Ph.D. Candidate in Computer Science	
Case Western Reserve University	2020 - 2022
• M.S. in Computer Engineering	Course-Focused Track
University of California, Irvine	2016 - 2020
• B.S. in Computer Science	Intelligent System Specialization

Area of Interest

Wearable Devices, Internet-of-Things, Digital Health, Sport Analytics, Human-Computer Interaction, Machine Learning, LLM-Aided Designs.

PROFESSIONAL EXPERIENCE

VivoMetria LLC.	May 2025-Aug 2025
<i>Hardware Intern</i>	
Karpos Cultivation LLC.	Dec 2020-Aug 2022
<i>Co-Founder & Lead Hardware Engineer</i>	

PUBLICATIONS

iRing: Rich Interaction on Smart Ring with Multi-point Capacitive Touch Screen	
<i>Runze Liu, Taiting Lu, Zhenghao Li, Yixi Wang, Shengming Yuan, Mahanth Gowda</i>	ACM UIST, 2026
AquaTag: An Open-Source Wearable Platform Enabling Flexible Sensor Placement for Aquatic Sensing	
<i>Zhenghao Li†, Runze Liu†, Taiting Lu, Kaiyuan Lin, Yuxin Tian, Yijie Li, Hsueh-Hung Cheng, Akshi Kartik, Yicheng Jin, Mahanth Gowda</i>	ACM IMWUT, 2026
SmartDampener: An Open Source Platform for Sport Analytics in Tennis	
<i>Runze Liu, Taiting Lu, Shengming Yuan, Hao Zhou, Mahanth Gowda</i>	ACM IMWUT, 2024
Demonstration of VisRing: A Display-Extended Smartring for Nano Visualizations	
<i>Runze Liu, Christian Krauter, Taiting Lu, Mara Schulte, Alexander Achberger, Tanja Blascheck, Michael Sedlmair, Mahanth Gowda</i>	ACM UIST Adjunct, 2025
VisRing: A Display-Extended Smartring for Nano Visualizations	
<i>Taiting Lu, Christian Krauter, Runze Liu, Mara Schulte, Alexander Achberger, Tanja Blascheck, Michael Sedlmair, Mahanth Gowda</i>	ACM UIST, 2025
FuSenseRing: An Open-Source Platform for Robust Cuffless Blood Pressure Monitoring via Multimodal Sensor Fusion and Temperature-Adaptive Attention	
<i>Zhenghao Li, Taiting Lu, Runze Liu, Shengming Yuan, Jigar Gosalia, Christine Bowlus, David Proctor, James Pawelczyk, Yincheng Jin, Mahanth Gowda</i>	ACM UIST, 2025
One Ring to Rule Them All: An Open Source Smartring Platform for Finger Motion Analytics and	

Healthcare Applications

Hao Zhou, Taiting Lu, Yilin Liu, Shijia Zhang, **Runze Liu**, Mahanth Gowda. ACM/IEEE IoTDI, 2023

I am an Earphone and I can Hear my Users Face: Facial Landmark Tracking using Smart Earphones

Shijia Zhang, Taiting Lu, Hao Zhou, Yilin Liu, **Runze Liu**, Mahanth Gowda ACM TIOT, 2023

LMM4-IC8K: A Large Multimodal Model Powered Integrated Circuit Footprint Geometry Understanding

Yida Wang, Taiting Lu, **Runze Liu**, Lanqing Yang, Zhe Chen, Yuehai Wang, Yixin Liu, Kaiyuan Lin, XiaoMeng Chen, Dian Ding, Yijie Li, Yifan Yang, Yi-Chao Chen, Yincheng Jin, Mahanth Gowda ICML 2026

OmniMech: All-in-one Multimodal Mechanical Benchmark for 3D Reconstruction

Taiting Lu†, **Runze Liu**†, Ziwei Dong, Sisong Bei, Jingying Zeng, Mingjia Wang, Zhenghao Li, Kaiyuan Lin, Yi-Shan Wu, Yangshoudu Zheng, Hongxing Pan, Kai Zhang, Guoliang Shi, Ling Ma, Yifan Yang, Jiaying Lu, Qi He, Sung-Liang Chen, Yi-Chao Chen, Yincheng Jin, Mahanth Gowda Preprint 2026

OmniRouting: A Semantic-Coupled Multimodal Benchmark for Constraint-Aware Spatial Reasoning in PCB Routing

Taiting Lu, Kaiyuan Lin, Ziwei Dong, Sisong Bei, Haolin Ye, Yuxin Tian, **Runze Liu**, Mingjia Wang, Jingying Zeng, Hongxing Pan, Kai Zhang, Haoyu Wang, Guoliang Shi, Ling Ma, Yifan Yang, Jiaying Lu, Qi He, Yi-Chao Chen, Sung-Liang Chen, Yincheng Jin, Mahanth Gowda Preprint 2026

OmniLayout: A Schematic-Coupled Multimodal Benchmark for Constraint-Aware Geometric Reasoning in PCB Layout

Taiting Lu, Kaiyuan Lin, Mingjia Wang, Haolin Ye, **Runze Liu**, Yuxin Tian, Vahe Melkonyan, Haoyu Wang, Muchuan Wang, Chufan Hong, Yifan Yang, Sung-Liang Chen, Yi-Chao Chen, Yicheng Jin, Mahanth Gowda Preprint 2026

ACTIVITIES AND REWARDS

NSF National I-Corps Program Cohort 3 Winter 2026

I-Corps Mid-Atlantic Regional Program Fall 2024

Best Paper Award for Edge IoT IoTDI 2023

PATENT

Apparatus, Process and Platform for Activity Analysis U.S. Prov. Pat. App. 63/688,633

Platform for Finger Motion Analytics and Healthcare Applications WO2024/233526

Automatic Heating and Temperature-controlled Insulated Bottle ZL2015/2/0187057.7

MENTORED STUDENTS

Yixi Wang

Undergraduate, Pennsylvania State University 2025-

Hsueh-hung Cheng

Undergraduate, Pennsylvania State University 2025-

Shengming Yuan

Undergraduate, Pennsylvania State University 2023-2024